

Executive Summary

Predicting Outpatient Consultation Duration — Hangu Outpatient Oncology Clinic

Ariana Butina

2026-07-02

The Question

How long will a given outpatient consultation take — and can that be predicted well enough, using only information already known at booking, to plan clinic capacity and estimate the true cost of a visit?

What We Built and Found

Using two years of real visit records (6,637 consultations), we built two tools that use only booking-time details — whether the patient has visited before, the day and time of the appointment, and whether a cancer diagnosis is on file — to (1) flag whether a visit is likely to run long, and (2) estimate its duration in minutes.

- **The “long visit” flag is moderately reliable** — correct about 6 times out of 10, clearly better than guessing but not precise enough to be the only scheduling factor.
- **The duration estimate is directionally useful, not exact** — typically within a little over four minutes on average, enough for capacity planning, not for setting one slot with confidence.
- **The single biggest driver of visit length is whether the patient has been seen before** — far ahead of the day of week, time of day, or diagnosis on record.

What We Recommend

- **Scheduling:** use these estimates to plan overall daily/weekly capacity and staffing, not to fix the exact length of any one appointment.
- **Costing (Time-Driven Activity-Based Costing):** segment cost-driver time estimates by **new vs. returning patient** — this carries far more real signal than segmenting by day, time slot, or diagnosis category.
- **Next step:** capture a little more clinical detail (reason for visit, complexity) to improve accuracy — scheduling data alone has a real but limited ceiling.

Bottom Line

Ordinary scheduling data is enough to produce a genuinely useful, if imperfect, estimate of consultation length and to flag visits at risk of running long — enough to support smarter capacity planning and a more evidence-based costing model today, as a planning aid for administrators, not a substitute for clinical judgment on any individual visit.